

EfficientNet Convolutional Neural Network for Automated Cervical Spine Fracture Identification

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ABSTRACT: The aim of this study is to develop an automatic system that uses deep learning techniques to detect cervical spine fractures from medical images. Using a DICOM (Digital imaging and communications in medicine) images of cervical spine, the system can identify a fracture and specify which vertebrae has a fracture. This process involves first processing provided DICOM images, then applying data augmentation techniques including metadata information, finally designing neural network architecture for feature extraction. We use a method called transfer learning, utilizing a pre-trained model like EfficientNet, which is used for feature extraction, while optimizing network parameters using AdanW optimizer and CosineAnnealingWarmup learning rate scheduler. The results demonstrate the successful training of neural network models to identify fracture and that fractured vertebrae, with commendable performance and accuracy level noted during evaluation on validation data. This study has important implications for medical imaging and healthcare industries, since automated fracture detection can increase diagnostic efficiency and accuracy. Moreover, such a system would provide good support to healthcare professionals, specifically radiologists, to identify cervical spine fracture easily.

Keywords: Cervical Spine Fracture, Classification, Convolution Neural Network(CNN), Deep Learning, DICOM, EfficientNet.

I. INTRODUCTION

The cervical spine is the uppermost part of the vertebral column and located in the neck area. Vertebral column made up of seven vertebrae labelled as C1 to C7, commencing at the base of the skull and culminating with thoracic spine. This contain axis, atlas and occiput with a chain of joints with three degrees of freedom and three axes

[1].The cervical spine is divided into the upper cervical spine (occiput-C2) and the lower cervical spine (C3-C7) [2]. It plays a vital role in supporting a head and allowing a body to perform different movements such as left and right axial rotation, as well as lateral bending, extension and flexion [3].

Cervical spine fractures are usually caused by car accidents, sports injuries, diving into shallow water or gunshot wounds [4]. However, these injuries cause severe complications such as spinal nerve or cord injury, postoperative infection, defective consolidation and in rare cases, lead poisoning from gunshot wounds and even paralysis if not properly diagnosed [2]. Current notable challenges in emergency medicine and trauma care demanding timely and accurate detection of fracture for effective treatment planning and patient recovery. Hence, Diagnostic images play a vital role for spine related diseases treatments [5].

Traditionally doctors examine cervical spine fractures manually where the process involves the doctor asking several questions to the patient related to injury. followed by, physically examining nerve problems. Further, doctors use stiff collars or stiff braces to stabilize the neck to prevent movement and further injury. Finally, they take a CT scan picture to determine appropriate diagnosis and manually take a decision which treatment is required [6]. This suggests that Radiologists are basically highly rely on manual methods. Diagnosing the patient based on personal opinion and process required a great amount of time. However, with advancements in Deep Learning (DL), there is an opportunity to detect a fracture using an automatic method.

Currently, we have seen that deep learning based models, specifically Convolutional Neural

Networks (CNN) models, are used in diverse healthcare fields, such as Radiology [7], Ophthalmology [8] and Dermatology [9]. Previously studies mostly focused on brain diseases as compared to spinal injuries because MRI successfully diagnoses brain related diseases.

This study proposes to use DL to find cervical spine fractures because it showed good results in the medical area. One of the DL algorithms EfficientNet developed based on the width, resolution and scaling of depth demonstrates promising results in various image classification tasks. Purpose of this study is to develop an automatic, robust and more accurate model to enable radiologists to streamline diagnoses of injuries in emergency situations. This research proposes a new method that uses deep learning and integrates metadata to enhance the performance. For comprehensive analysis, image features need to be integrated with additional data such as thickness and location of the slice. This will enhance the understanding regarding images and identify cervical spine fractures more precisely. Study involves identification of vertebrae and vertebral fracture by implementing EfficientNet. Our experiments show that the EfficientNet algorithm results in high accuracy, precision, recall, and F1 scores. Primarily, our aim is to develop a robust model to accurately identify cervical spine fractures. Therefore, we investigated the feasibility of using convolutional neural networks (CNN) with a pre-trained model EfficientNet (to extract features) to identify cervical spine vertebrae and fracture.

II. RELATED WORK

Naguib et al. [10] implemented a computer-aided-diagnosis system using deep learning models such as, AlexNet and GoogleNet to classify cervical spine injuries as fractures or dislocations. They utilized saliency maps to highlight important areas in X-ray images, achieving high accuracy rates and aiming to improve the efficiency of cervical spine injury diagnosis.

Hojjat et al. [11] implemented neural network with CT scan images. The performance was measured with balanced and imbalanced dataset.

Small et al. [12] implemented a CNN to detect cervical spine fractures on CT scans. Model was trained on different datasets from multiple organizations and CT scanners, including validation which showed high sensitivity and specificity. This tool aims to assist radiologists in efficiently identifying cervical spine fractures.

Golla [13] implemented neural network using tomographic image data. U-net was used to create the segmented images.

Nicola et al. [14] implemented 3D CNN for automatic vertebrae segmentation and k-means clustering for vertebrae localization using traditional machine learning, known as a two-fold approach. The integration of deep learning and traditional algorithms enhances the efficiency and accuracy of spine analysis in CT scans.

III. METHODOLOGY

a. Dataset

The dataset provided comes from the collaboration between the Radiological Society of North America (RSNA), the American Society of Neuroradiology (ASNR), and the American Society of Spine Radiology (ASSR) [15]. It consists of four distinct CSV files and three directories containing scan slices and segmentations for a total of 87 related files.

Dataset reveals a fair balanced distribution of target sites, with a split of 52/48. This suggests an equal spread of the samples between the two output categories. Interestingly, C7 stands out with the highest proportion of fractures at 19%, while C3 exhibits the lowest at 4%. This difference suggests that certain cervical vertebrae may be more prone to fractures than others.

Most of the samples are sized at 512 x 512 pixels. There are also 245 images with dimensions of 512 x 519 pixels and 782 images that are 768 x 768 pixels. To keep things consistent for the CNN input data, the remaining images are resized to 512x512 pixels.

Getting data ready for deep learning models involves a few important steps, like pre-processing and image augmentation [16]. During augmentation different image parameters such as brightness, contrast, cropping is adjusted to fit the model appropriately. Image augmentation, on the other hand, is like giving the dataset a makeover, where copies of the images were tweaked, rotated, flipped, and zoomed as necessary. This helps the model to learn from a range of examples, making it better at handling different situations. Augmentation mainly helps in generalizing the model to handle new types of samples. Detecting cervical spinal fractures involves two main steps:

- Identifying the vertebrae number in the current slice.
- Determining whether a fracture is present or not.

To achieve this, two distinct models using EfficientNet V2 were created as backbone.

b. EfficientNetV2

EfficientNetV2 also comes with some fresh design features, like inverted residual blocks and linear bottlenecks. Those are clever solutions to

shrink down the model size and cut down on computing power, all without sacrificing performance. This makes it perfect for running on devices and platforms that don't have a ton of resources to spare. Using EfficientNet it is possible to adjust the width, resolution and depth using some scaling factors.

c. Vertebrae Detection

Initially, DICOM [17] images are processed and used as an input to obtained vertebrae numbers as an output from NIFTI [18]. EfficientNet is used as a backbone network to enhance the accuracy with minimal parameters. This significantly improves training and predication efficiency.

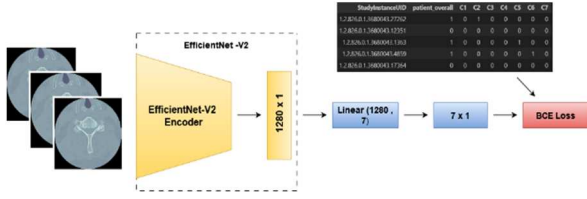


Fig 1: Architecture of Vertebrae Detection Model

The model architecture is illustrated in figure 1. Initially input the DICOM files to the EfficientNet V2 model extract essential image features. Subsequently, mathematical transformations are applied to reduce dimensionality. Extracted features represent the probability of specific vertebrae, enabling vertebrae identification. To improve the model performance a custom Binary Cross Entropy loss function can be used. It compares the model's predictions with expected output and improves model performance.

d. Fracture Detection

Second model in Figure 2 illustrates fracture detection process from the DICOM slices. To ensure reliability, as an input DICOM image and extract important metadata as shown in figure 2 are incorporated into the model.

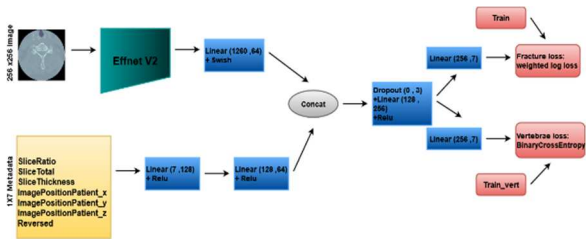


Fig 2 : Architecture of Fracture Detection Model

DICOM images are processed through EfficientNet V2 to extracted key features (tensor sized 1280x64). Concatenate with metadata. A dropout layer is incorporated to prevent any individual model component from dominating the learning process. Loss functions are employed to measure performance for identification of areas for

improvement. For instance, Binary Cross Entropy for vertebrae detection and Custom Log for Fracture detection.

e. Optimizer

The AdamW optimizer [19] resembles the Adam optimizer. It integrates weight decay directly into its update step to mitigate the risk of excessive weight during training. This modification enhances stability of training and performance, especially in deep learning. Using AdamW, neural network training becomes more effective overall by reducing overfitting and enhancing the model's capacity to process new data.

f. Learning Rate Scheduler

Cosine annealing warmup restarts [20] learning scheduler functions as a kind of intelligent training guidance for deep neural networks. As training goes on, it gradually reduces the learning rate to keep the model from being trapped on less-than-ideal solutions. It assists the model in exploring various pathways and enhances overall performance by regularly resetting the learning rate. In order to help the model navigate training more effectively and produce better training outcomes and greater generalization to new data, this scheduler modifies the learning rate based on the cosine function.

g. Dropout

Dropout is a regularization technique used in neural networks. Regularizing is one of the important features to mitigate the risk of overfitting during the training. Specifically in Convolution neural networks (CNNs), a dropout layer is applied in fully connected layers [21]. Meanwhile, it also improves generalization by avoiding over reliance on specific neurons leading to model robustness. This is particularly beneficial for medical imaging, where slight variation can occur.

h. Model Evaluation

The proposed system performance is checked with validation dataset using different parameters such as accuracy, precision, recall, and F1 score are used [22]. The loss function used is shown in equation 1. The equation helps to find the probability of vertebra having fracture or not.

$$L_{ij} = -w_j(y_{ij}\log(p_{ij}) + (1 - y_{ij})\log(1 - p_{ij})) \quad (1)$$

IV. RESULT

a. EfficientNet Vertebrae Detection

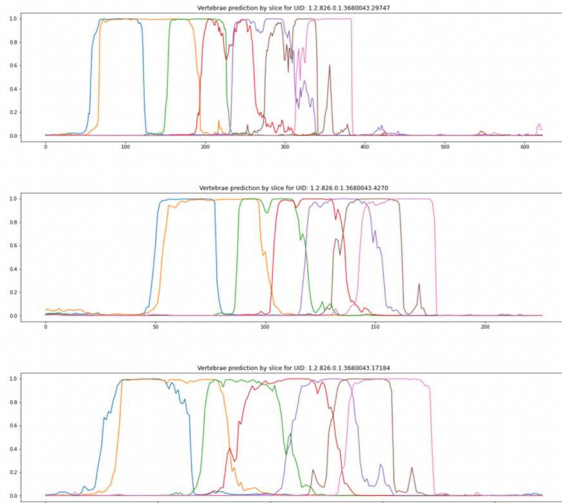


Fig 3 : Vertebrae prediction by slices for 3 patients

In figure 3, shows the performance of the algorithm with its segmentation task, while predicting the vertebrae for 3 patients. Each colour on the image represents a different vertebra.

b. EfficientNet Fracture Detection

Figures 4 and 5 illustrate the performance measure of the model as per the fracture and vertebrae prediction of the patient. Figure 8 indicates the high possibility of a fracture in the C7. It indicates the presence of fracture in any vertebrae. The “Patient Overall” column indicates overall vertebral fracture status.

vertebrae. Additionally, it also accurately distinguishes between different vertebrae and assigns a distinct score to each slice.

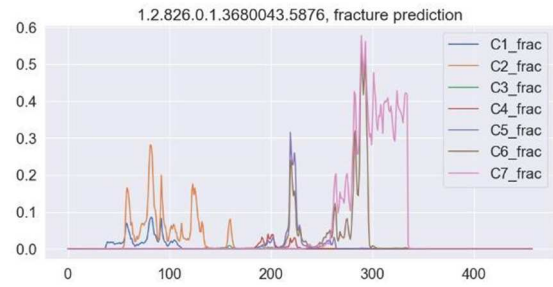


Fig 4 : Weights for fracture prediction

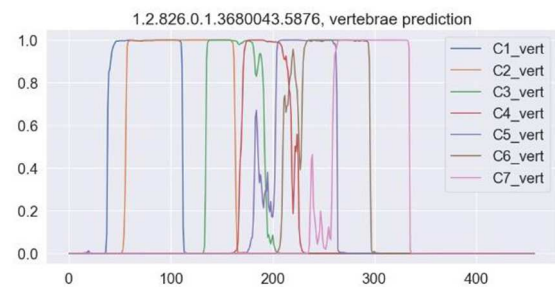


Fig 5 : Weights for vertebrae prediction

Using the Cosine Annealing Warmup Restarts scheduler, dynamically modify the learning rate and enhance both the convergence and overall performance of the model. Moreover, testing the model on several folds validated its robustness and good generalization to fresh data. Table 1 shows the prediction of one patient.

Table 1 : Final Prediction on Test Data

S_UID	Patient_Overall	C1	C2	C3	C4	C5	C6	C7
1.2.826.0.1.3680043.10400	1.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
1.2.826.0.1.3680043.10576	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
1.2.826.0.1.3680043.10628	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
1.2.826.0.1.3680043.10678	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0

Fracture and vertebrae prediction model is performing admirably, boasting an average accuracy of approximately 91.2%.

Table 2: Accuracy for each vertebrae Class

Class	Accuracy	Precision	Recall	F1-Score
C1	0.9024	1.000	0.902	0.9487
C2	0.8780	0.972	0.900	0.9350
C3	0.9268	1.000	0.926	0.9620
C4	0.9024	1.000	0.902	0.9487
C5	0.9512	1.000	0.951	0.9750
C6	0.4390	0.416	0.882	0.5660
C7	0.6097	0.628	0.880	0.7333

Classes such as, C1, C3, C4, and C5 have F1-scores above 94% and a good precision and recall score. C6 and C7 witnesses a lower accuracy because of data imbalance. The concept has potential to benefit medical practitioners, notwithstanding certain difficulties. We're dedicated to keeping it improved so that practical medical uses might benefit from even greater efficacy.

V. CONCLUSION

The study highlights the potential of using deep learning approaches to automatically detect cervical spine fractures. The EfficientNet model shows promise in enhancing diagnostic accuracy and speeding up treatment for patients with cervical spine injuries. Moving forward, it would be valuable to fine-tune the model's design and integrate it into clinical setups to improve patient care and healthcare efficiency.

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